

RoadWatch

AI-Powered Road Safety & Transparency Platform — connecting citizens, government authorities, and contractors into one unified ecosystem to Report, Track, Verify and Improve roads across India.

"Report → Track → Verify → Improve"

5 Phases	100% Free Stack	Aadhaar + DigiLocker	Gemini Free Tier	PWA Offline	Open Source GIS
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■ ZERO-COST ARCHITECTURE

Every tool, API, and service in this PRD uses a free tier or is open-source. No credit card required. No paid subscriptions. Designed for hackathon build and beyond.

1. PROBLEM STATEMENT

No visibility into spending	Citizens cannot verify where public road funds go or check if spend matches actual road quality.
No structured feedback channel	No trackable way to report potholes, cracks, waterlogging, open manholes, broken dividers, or missing lights.
No accountability	Authorities face no public pressure to act. Complaints disappear without follow-up.
Fake reports & spam	Anonymous, unverified reporting enables abuse and dilutes priority signals.
No repair verification	Tickets are self-closed by authorities with no independent community confirmation.
Complex government data	Tender documents are 40+ pages; average citizens cannot audit contractor appointments or budgets.
Contractor impunity	No public record of contractor performance, re-repair rates, or road durability scores.

2. TARGET USERS

Role	Goal	Pain Point
Aadhaar-verified citizen	Report issues, upvote existing reports, track status, verify completed repairs	No confirmation repairs are actually done after reporting
Anonymous whistleblower	Report corruption, negligence, or unsafe infrastructure without identity exposure	Fear of retaliation when filing formal complaints
Municipal authority	Triage reports by AI priority, assign contractors, close verified tickets	Fragmented complaints across phone, email, social media
Contractor / PWD team	Receive work orders, update progress, upload before/after photo proof	No clear ownership or public accountability per job
Researcher / Journalist / NGO	Audit budget spend vs verified repair quality via Open Data API	Spending and road quality data live in separate government silos

3. IDENTITY & AUTHENTICATION (FREE)

Aadhaar eKYC + DigiLocker: One Aadhaar = one RoadWatch account. Prevents spam, ensures upvotes reflect real citizens, and makes the verification loop trustworthy. Both UIDAI eKYC and DigiLocker APIs are **free for registered applications** — use the UIDAI sandbox for the hackathon demo.

Step 1 User enters mobile number linked to Aadhaar	Step 2 UIDAI sends OTP to registered mobile (free sandbox)	Step 3 OTP verified; eKYC returns name + district + ward	Step 4 Unique account created — Aadhaar hash stored, number never retained
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DigiLocker: Optional secondary login for users who prefer it. Shares verified name and state — no address or document data. **Anonymous mode** also available for corruption/negligence reports — no login required but limited to 1 anonymous report per device per day (IP rate-limited).

4. GEMINI AI INTEGRATION (FREE TIER — GEMINI 1.5 FLASH)

Gemini 1.5 Flash via Google AI Studio is **completely free** (15 RPM, 1M tokens/day — sufficient for a hackathon and early-stage deployment). Powers five intelligent layers:

Gemini Feature	How it works	Free API
Location Intelligence	User types a road name or uploads a photo. Gemini Vision identifies landmarks and road name, cross-references OpenStreetMap geocoding, auto-fills ward, district, PIN, managing authority (NHAI / PWD / Municipal), contractor name, budget allocation, last maintenance date, and tender summary.	Gemini 1.5 Flash + Nominatim OSM
AI Damage Classification	Analyses uploaded road photo. Returns: issue type (pothole / crack / waterlogging / open manhole / broken divider / missing streetlight / faded zebra crossing), severity (Low / Medium / Critical), and estimated repair urgency score.	Gemini Vision Free tier
Tender Summariser	Govt tender PDF uploaded → Gemini returns 4-bullet plain-language summary: what work, which road, which contractor, what cost, deadline. Enables citizen auditing of 40-page documents.	Gemini 1.5 Flash Free tier
Cost Estimator	Compares government-declared spend per km against Gemini-estimated real market repair cost. Flags overspending or potential corruption with a percentage deviation score.	Gemini 1.5 Flash Free tier
AI Citizen Chatbot	In-app assistant answers: 'Who manages this road?', 'What is the status of my complaint?', 'How do I escalate?'. Responds in Hindi and English.	Gemini 1.5 Flash Free tier

5. COMPLETE FEATURE REQUIREMENTS

Feature	Description	Pri	Owner
Aadhaar + DigiLocker login	UIDAI OTP eKYC + DigiLocker. One account per Aadhaar. Hash stored, number never retained. Free sandbox for demo.	P0	Backend
Anonymous reporting mode	Report corruption/negligence without login. Rate-limited to 1/device/day via IP hashing. Identity never stored.	P0	Backend
Smart interactive map	Leaflet.js + OpenStreetMap (100% free). GPS auto-detect or manual pin drag. Live complaint markers with clustering.	P0	Frontend

Geotagged photo + video upload	EXIF GPS extraction from uploaded photos auto-pins location on map. Timestamp captured. Video upload supported.	P0	Frontend
AI damage classification	Gemini Vision classifies: pothole / crack / waterlogging / open manhole / broken divider / missing light / faded crossing. Severity: Low / Medium / Critical.	P0	AI
Work-order lifecycle	Public 5-step status bar: Reported → Under Review → Assigned to Contractor → Work in Progress → Resolved. Visible to all citizens.	P0	Backend
Complaint registration	Description, photo/video, auto-location, issue category. Auto-routed to responsible authority by ward.	P0	Backend
Gemini location intelligence	Road name or photo → Gemini fetches managing authority, contractor, budget, last maintenance date, approval info via OSM + public data.	P0	AI
Dynamic heatmaps	Accident-prone zones + high complaint density = Danger Zone heatmap layers. Helps authorities prioritise repairs.	P1	Frontend+Data
Verification loop	Before ticket closes, nearby Aadhaar-verified citizens prompted to upload verification photo. Gemini confirms repair visible. Requires 2 positive verifications to close.	P1	Backend+AI
Live expenditure map	Active road projects, repair zones, construction sites shown on map. Click road → see budget, contractor, deadline, status.	P1	Frontend+Data
Contractor performance profile	Public profile: completed projects, citizen ratings, failure rate, road durability score. Auto-decrements if repaired road re-damages within 3 months.	P1	Backend
Contractor leaderboard	Public ranking by resolution speed, re-open rate, citizen verification score. Fully public — no login required to view.	P1	Frontend
Tender summariser	Upload govt PDF → Gemini returns 4-bullet plain-language summary of what, where, who, how much, by when.	P1	AI
Cost estimator tool	AI compares declared govt spend vs estimated market cost. Flags overspending percentage. Helps identify corruption.	P1	AI+Data
Upvote / support system	Aadhaar-verified upvotes on existing complaints. More upvotes = higher priority + deeper heatmap intensity.	P1	Frontend
AI citizen chatbot	Gemini-powered assistant. Answers road authority questions, complaint status, escalation steps. Hindi + English.	P1	AI
Real-time safety alerts	Web Push (free) notifications for dangerous roads, accident-prone zones, flooded roads, ongoing construction nearby.	P2	Backend
Nearby emergency services	Map overlay for hospitals, police stations, ambulance services, vehicle recovery near reported location.	P2	Frontend

3D road visualisation	Three.js digital twin of damaged road section. Degrades to 2D photo on low-end devices (WebGL check).	P2	Frontend
Open Data API	Public CSV/JSON exports of road quality data, complaint analytics, spending data for researchers and journalists.	P2	Backend
Public transparency dashboard	Total / resolved / pending complaints. Dept performance analytics. Budget utilisation. Contractor rankings. No login required.	P2	Frontend
PWA + offline mode	Installable PWA via Web App Manifest. Service Worker queues reports offline using IndexedDB. Syncs on reconnect.	P2	Frontend

6. WORK-ORDER LIFECYCLE DETAIL

1. Reported Aadhaar citizen or anonymous user submits. Gemini classifies + auto-routes to authority.	2. Under Review Authority views heatmap context, confirms AI priority, accepts or reassigns.	3. Assigned Work order generated. Contractor visible on public leaderboard. Citizen notified via Web Push.	4. Work in Progress Contractor uploads progress photos. Status advances publicly in real time.	5. Resolved ✓ 2 nearby citizens upload verification photos. Gemini confirms fix. Ticket closes. Contractor score updates.
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7. FREE TECHNOLOGY STACK (ZERO PAID SUBSCRIPTIONS)

Every tool listed below is either **open-source**, has a **permanent free tier**, or is **free for hackathon/development use**. No credit card required for any of them.

Layer	Free Tool / Service	Why Free	Paid Alternative Avoided
Identity	UIDAI eKYC Sandbox + DigiLocker API	Free for registered apps; sandbox free for dev/KYC providers	Paid KYC providers
AI Core	Gemini 1.5 Flash — Google AI Studio	Free: 15 RPM, 1M tokens/day, no credit card	OpenAI GPT-4 (paid)
Maps / GIS	Leaflet.js + OpenStreetMap (Nominatim)	100% open-source and free, no API key reqd	Mapbox GL JS (paid beyond free tier)
Geocoding	Nominatim (OSM) / OpenCage free tier	2,500 requests/day free on OpenCage; Nominatim unlimited	Google Maps Geocoding API (paid)
3D Twin	Three.js	Open-source, no usage limits	Unity / commercial 3D SDKs
Frontend	React + Tailwind CSS	Both fully open-source	Paid UI component libraries
PWA	Service Worker + IndexedDB (Web App Manifest)	Built into every modern browser, zero cost	Paid app distribution
Backend	Node.js + Express	Open-source, free to self-host	Commercial BaaS platforms
Database	Supabase free tier (PostgreSQL + PostGIS)	Free: 500MB DB, 1GB storage, 2GB bandwidth	AWS RDS / PlanetScale paid
File Storage	Supabase Storage (free tier)	1GB included in Supabase free tier	Cloudinary paid / AWS S3
Auth (extra)	Supabase Auth (free)	Included in free tier — handles JWT sessions	Auth0 paid
Notifications	Web Push API + Firebase FCM (free)	FCM is free with no message limits	Twilio SMS (paid)
Hosting FE	Vercel (free tier)	Free for hobby/hackathon projects	AWS CloudFront (paid)
Hosting BE	Render free tier	Free tier: 750 hrs/month	AWS EC2 (paid)
PDF parsing	pdf-parse (npm, open-source)	Open-source, no limits	Adobe PDF API (paid)
Charts	Chart.js / Recharts	Open-source	Highcharts (paid)

8. IMPLEMENTATION ROADMAP (5 PHASES — HACKATHON + BEYOND)

Phase	Name	Key Features	Hours
Phase 1	Capture	Aadhaar eKYC login, anonymous mode, Leaflet map with GPS pin, geotagged photo upload, EXIF extraction, basic complaint form, auto-route to ward authority	0–12h
Phase 2	Analyse	Gemini damage classification, location intelligence (authority + contractor fetch), dynamic heatmaps, AI priority scoring, work-order status bar, spending overlay with mock data	12–28h
Phase 3	Account	Budget tracking by ward, contractor performance profiles, leaderboard, tender PDF summariser, cost estimator vs market price, AI citizen chatbot, authority dashboard	28–40h
Phase 4	Close	Verification loop (2-citizen photo confirm + Gemini AI validation), upvote system, Web Push notifications, real-time safety alerts, nearby emergency services overlay	40–48h
Phase 5	Scale	PWA offline mode + Service Worker, 3D road twin (Three.js), Open Data API (CSV/JSON), public transparency dashboard, multilingual support (Hindi + English)	Post-hackathon

9. RISKS & MITIGATIONS

Risk	Sev	Mitigation
UIDAI sandbox limits during peak demo	High	Pre-seed 5–10 test Aadhaar accounts in sandbox. Demo uses pre-verified sessions
Gemini free tier rate limit (15 RPM)	High	Queue classification requests. Cache identical photo results by hash. For demo, pre-p
No real government spending data available	Medium	Mock ward-level budget CSV with realistic figures. Clearly labelled 'Sample Data'. In p
GPS/EXIF missing from screenshots or iOS photos	Medium	Fall back to manual pin drag on map. Gemini landmark recognition as secondary loca
Supabase free tier 500MB DB limit	Medium	Compress photos before upload (client-side canvas resize to max 800px). Estimated
Citizen verification fatigue	Low	Limit verification prompts to users within 500m. Gamify with 'Verified Citizen' badge. M
3D twin performance on low-end Android	Low	Detect WebGL support on load. Gracefully degrade to 2D before/after photo comparis
Contractor disputes public rating	Low	Rating changes only on algorithmic triggers (re-damage < 3 months). Authority can fla

10. SUCCESS METRICS

Metric	Hackathon Target	Production Target
Verified report submissions	10 demo submissions	> 100 reports/day per pilot ward
Duplicate account attempts blocked	0 during demo	0 (Aadhaar hash collision check)
Gemini classification accuracy	Validated on 20 test photos	> 85% match with manual review
Average work-order resolution time	N/A (simulated in demo)	< 7 days for Critical issues
Citizen verification completion	End-to-end shown in demo script	> 60% of closed tickets verified
Spending overlay accuracy	Mock data with realistic structure	Matches RTI data within 5%
Contractor leaderboard views	Shown in demo	> 40% of weekly active users
Tender summary reads	1 tender summarised live	> 30% of uploaded tenders get views

11. HACKATHON DEMO SCRIPT (90 SECONDS)

#	Action	What the Judge Sees
1	Open RoadWatch — citizen logs in with Aadhaar OTP flow; name + ward auto-populate from eKYC. No manual form.	
2	Upload a pothole photo from camera roll	Gemini classifies: Pothole / Critical severity. Location auto-fills: 'NH-58 Meerut Highway — main road'.
3	Submit complaint	Pin appears on Leaflet map inside a red Danger Zone heatmap cluster. Upvotes from nearby users.
4	Switch to Authority Dashboard	New ticket visible — AI priority HIGH. One-click assigns contractor. Work order generated. Status: Pending.
5	Contractor updates to 'Work in Progress'	Status bar advances publicly. Citizen receives Web Push notification in browser.
6	Contractor marks Resolved — uploads after photo	System prompts 2 nearby verified citizens to upload verification photos.
7	Citizen uploads verification photo	Gemini confirms repair visible → ticket turns green on map. Contractor score updates on leaderboard.
8	Open Tender Summariser tab — upload a PDF	Gemini returns 4-bullet summary in 3 seconds. Judge can see contract details without reading PDF.

12. FINAL VISION STATEMENT

RoadWatch transforms citizens from passive commuters into active infrastructure auditors. By combining the trust of Aadhaar-based identity with the intelligence of Gemini AI and the accessibility of OpenStreetMap, we create a transparent digital layer over India's physical roads. Our goal is to turn every smartphone into a tool for governance, ensuring that public funds lead to safer journeys and that no pothole—or budget leak—goes unnoticed.